Question	Answer	Marks	Partial Marks
1	5.5 or $5\frac{1}{2}$ or $\frac{11}{2}$	**2**	**M1** for a correct first step $4c = 13 + 9$ or $c - \frac{9}{4} = \frac{13}{4}$ or better
2	8.0 cao	**2**	**B1** for 7.99[9…] or for *their* answer seen to 3 or more sf and rounded correctly
3	UK by [$]0.69 or [$]0.692…	**3**	**B2** for 1.96[8…] or 1.97 **and** 1.27[6…] or 1.28 or **M2** for $1.62 \times 1.215 - \frac{4.83}{3.785}$ oe or **M1** for 1.62 × 1.215 or $\frac{4.83}{3.785}$
4(a)(i)	Translation $\begin{pmatrix} 8 \\ -3 \end{pmatrix}$	**2**	**B1** for translation or for correct vector oe
4(a)(ii)	Rotation 90° [anticlockwise] [centre] (0, 0)	**3**	**B1** for each
4(b)	Image drawn at (0, 5) (0, 6) (2, 6) (2, 2) (1, 2) (1, 5)	**2**	**B1** for reflection in $x = k$ If 0 scored, **SC1** for reflection in $y = -2$
5(a)	1000 or 10^3	**1**	
5(b)	$n^2 + 4$ final answer oe	**2**	**B1** for second difference = 2 or for answer as a quadratic expression or for a correct expression shown in the working and then spoilt
5(c)(i)	9	**1**	
5(c)(ii)	12	**2**	**B1** for 21
6	$15x^3 - 10x^2$ final answer	**2**	**B1** for $15x^3$ or $-10x^2$ or correct answer seen in working and spoilt
7	$-1.5 \leqslant y < 4$ final answer	**2**	**B1** for $-1.5 \leqslant y$ or $y < 4$
8(a)	1099	**2**	**M1** for $\frac{20}{100} \times 980$

Question	Answer	Marks	Partial Marks
8(b)	12.1 or 12.14…	**2FT**	FT for 2 marks for $\left(\frac{their\ 1099}{980}-1\right)\times 100$ oe correctly evaluated to at least 3sf **M1** for $\frac{their\ 1099-980}{980}[\times 100]$ or for $\left(\frac{their\ 1099}{980}-1\right)[\times 100]$ or for $\frac{their\ 1099-980}{980}\times 100\,[-100]$
9	475	**2**	**M1** for $x\times\frac{8}{100}=38$ or for $\frac{38}{8}\times 100$ oe
10(a)	1, 2, 3, 6, 9, 18	**2**	**B1** for a list with one error or one omission
10(b)	$(y + 5)(3 – x)$ final answer	**2**	**M1** for $y(3 – x) + 5(3 – x)$ or for $3(y + 5) – x(y + 5)$ or for correct answer seen and spoilt
10(c)	$x = 2$, $y = 13$ or $x = 1$, $y = 4$	**2**	FT *their* **(b)** for **M1** **M1** for *their* $3-x=a$ and *their* $y+5=b$ where $ab=18$ and a, b integers
11	3200	**3**	**M2** for $800 \div 50^2\ [\times 10^n]$ oe or **B1** for 50^2 or 800×100^2 If 0 scored, **SC1** for 160 000 as final answer
12	8.36 or 8.364…	**3**	**M2** for $\frac{\text{dist}}{12.5}=\sin 42$ oe or better or **M1** for recognition of shortest distance soi
13	$[\pm]\sqrt{\frac{A-w^2}{5}}$ oe final answer	**3**	**M1** for $A – w^2 = 5x^2$ or for $\frac{A}{5}=\frac{w^2}{5}+x^2$ **M1** for $x^2 = \ldots$ FT *their* first step **M1** for correct square root , FT *their* $x^2 =$ An incorrect answer scores a maximum of **M1 M1**